

FlexNav: Flexible Navigation and Exploration through Connected Runnable Zones

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ABSTRACT

Runners want to actively explore unknown environments without the fear of getting lost. We conducted a survey to better understand runners' needs and practices in this context. The survey results emphasized the interest in flexible exploration. To address this interest, we designed FlexNav, a system that supports exploratory running through flexible tours that link the best runnable zones in a neighbourhood. FlexNav provides adaptive navigation support enabling runners to follow such tours without continually getting disruptive directions. We tested it with runners to assess its usability. The results confirm the usefulness of the system and the users' preference for flexible tours over fully specified tours with turn-by-turn guidance. Our study highlights the subjective nature of runnable zones and the subtle balance between guidance and exploration.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Usability testing; Interaction design; Ubiquitous and mobile computing; Ubiquitous and mobile devices; Mobile devices.

KEYWORDS

Running, Route Recommendation, Pedestrian Navigation, Sports

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1 INTRODUCTION

Runners want to exercise and enjoy their run without the fear of getting lost, even in unknown environments. For this purpose, most existing systems provide access to tours previously ran and uploaded by other runners, and some generate new tours on request. To follow such tours, runners mostly rely on turn-by-turn navigation with permanent voice guidance, which is often perceived as unpleasant and disruptive [4]. Furthermore, runners often prefer to remain flexible, i.e., they do not want to strictly follow a specific tour from start to end, an aspect that is also discussed in the literature [2]. Our survey results (described in Section 3) confirm this behaviour.

Overall, existing solutions for running in unknown environments support runners in two opposite ways, either by providing fully pre-planned tours with turn-by-turn navigation [5][6][7] or, on the contrary, by providing no pre-planned tours, but decision support enabling runners to choose the direction themselves dynamically, during the run [2]. Our aim is to provide a solution that optimally combines both approaches, proposing partially specified tours and adaptive navigation support depending on the quality of the environment (as defined in Section 3.2, inspired by [14][31][32]). The idea is to guide the runner effectively towards good quality running zones, and then give them freedom to explore these zones. Our primary focus is on urban areas, where it is more challenging to find and compose pleasant itineraries. Our solution focuses on recreational runners who simply want to put on their shoes and run [30]. It is less appropriate for competitive runners who have specific training needs when preparing a performance and who therefore require running tours with precise characteristics. However, our solution can also have a broader usage, e.g., for other recreational sports like hiking, cycling, or touristic pedestrian tours.

Our work was inspired by the hypothesis that, when running in unknown environments, runners often first search and select a good running zone to explore close to their start location. For instance, when visiting a new city, they will typically look at a map and select a park located not too far from their hotel as a good running zone. Then, they will try to reach that zone by taking the shortest route so that they spend most of the time available in the zone itself, running and exploring it spontaneously, instead of strictly following a pre-planned itinerary. To better understand the practices and needs of runners when running in and exploring unknown environments we first conducted a survey. Considering the

survey results, we then devised a novel solution enabling runners to explore the best runnable zones in the neighbourhood. Our system generates running tours by selecting and linking a set of runnable zones through best path connections, but without specifying a path to follow within these zones. It lets runners explore each of these zones freely before moving on to the next one. It continuously monitors the run and ensures that the runner finally reaches but does not exceed the desired target distance. It provides adaptive navigation support as follows: in the connecting paths between the zones, the runner is fully guided using turn-by-turn instructions; in the zones, the runner gets decision support but no active guidance and only minimal alerts, thus limiting disruption. Our solution provides instructions primarily through audio complemented with visual and haptic feedback modalities.

Our research aims at answering the following questions: “What are runners’ exploratory habits in known and unknown areas?” and “Does our proposed concept satisfy the needs of runners who wish to explore unknown places?”. To investigate the first question, we designed a survey and collected responses from 139 participants about their running habits and behaviours. We then designed our concept based on the insights collected through the survey and implemented a smartwatch-based prototype to test our concept of runnable zones with adaptive navigation. We chose a smartwatch due to its lightweight physical properties and its regular use amongst runners [22][30][33]. We asked participants for feedback about the usability of our system and to compare it with fixed turn-by-turn navigation support. We also investigated and compared the participants’ spatial learning in the fixed and flexible parts of the tour, in which our system respectively provides strict turn-by-turn and minimal navigation instructions.

This paper is structured as follows: first, we discuss related work concerning the generation of running tours, approaches to provide flexible navigation support during running, and the most suitable feedback modalities. Then we discuss our survey and its findings. Afterwards, we explain FlexNav, our concept of flexible running tours and the corresponding navigation support. We describe the prototype we implemented and the experiment we carried out to test it with users. We finally report the results of the experiment and limitations.

With our work, we contribute to the growing field of HCI in sports and novel methods for pedestrian navigation. More specifically, our contribution is threefold. First, we carried out a survey and analysis of running habits and behaviours in unknown environments. Second, we propose a concept for running tours composed of fixed and flexible parts. Finally, we present a corresponding adaptive navigation support for flexible explorative running that adapts the level of guidance to the environment characteristics.

2 RELATED WORK

Most of the literature on the use of technology to support runners focuses on apps and devices monitoring performance. Nevertheless, existing work identifies a gap between runners’ expectations and their actual experience with these mobile sport applications [30]. Knaving et al. [34] state that when running, amateur runners primarily seek exploration and relaxation and that one of the key practicalities to their everyday run is “the possibility to easily vary

the length and difficulty of a run without having to plan beforehand”. These notions of exploration and flexibility are also part of the core findings of a study conducted by McGookin and Brewster [2]. Exploration means to discover unknown paths and places. Flexibility means to be able to dynamically adapt the run and the overall tour to an evolving context, such as changing weather or fatigue.

Two complementary approaches exist that support explorative and flexible running in unfamiliar environments. The first one focuses on guided exploration: it enables runners to practice their activity in unknown places by providing them with pleasant running tours that they can follow, typically using turn-by-turn navigation. The second one rather focuses on supporting flexibility during the run: it does not provide a fixed pre-planned tour but focuses on providing appropriate navigation decision support, enabling the user to dynamically choose a pleasant path to follow. Below we will discuss both approaches as well as the different feedback modalities that have been used in prior art to provide navigation support for runners and pedestrians in general.

2.1 Providing Pleasant Tours and Zones

To enable runners to explore new running paths in unknown environments, available commercial systems give access to existing tours or compose new running tours from user-generated content, i.e., existing collections of running traces [5][6][7]. Existing work has also explored the generation of running tours based on map characteristics [14] or the filtering and selection of the best existing itineraries using one [8] or more route characteristics [1][3]. However, all these approaches end up returning only one specific path to follow and discard all alternative path options that might be equally interesting. Thus, they do not enable any flexibility during the run. An approach to make these tours flexible would be to keep the discarded alternatives and let the runner decide, during the run, which of the options to follow.

Our approach to enable exploration and flexibility in the context of running is to determine and add homogeneous geographic zones of good running quality to the pre-planned tours. In other application domains, researchers have explored the implementation of virtual perimeters to delimit geographic zones of interest. This is known as geo-fencing [25] and has been used in location-based services to detect if a tracked person or object enters or leaves a specific area. This information is used to trigger notifications (e.g., for targeted advertisement [26]) or alerts (e.g., child or elderly location services that warn caregivers or family when they leave a designated area [27][28]). However, we are not aware of any previous work leveraging this concept to generate flexible running tours as we propose in our system.

2.2 Navigation Support providing Flexibility

Existing approaches to navigation support for pedestrians, cyclists, and runners consist mostly of turn-by-turn directions to efficiently guide the user along a given path. The need to provide more flexible navigation support has been identified in prior art [2], but there is little work [29] that explores such support. In the context of running, RunNav [2] supports explorative and flexible running through a smartwatch application. However, it does not provide

neither pre-defined running tours nor navigation support enabling runners to follow such tours. Its focus is rather on supporting fully flexible running by enabling runners to take all navigation decisions themselves, dynamically, during the run. Therefore, RunNav provides the runner with visual navigation decision support on the smartwatch. It displays an egocentric color-coded map indicating the estimated quality for running of the areas surrounding the runner. The runner then consults this map to decide on the path to take, based on the quality of the surrounding areas.

Prior art has also explored bearing-based haptic feedback for pedestrians to enable more flexible navigation towards a target destination. The system designed in [23] lets the user actively scan possible directions whenever they want by pointing their smartphone in those directions; in return, it provides information about the correct bearing and the number of alternative path options available for each direction. Other work [24][35] provide vibrotactile feedback about the bearing towards the destination through the wristband of a smartwatch that encodes direction information while allowing the user to decide on the concrete path to follow.

While the approaches discussed above enable exploration and flexibility, they also put a significant burden on the user making the decisions. While this burden is limited in overall good quality surroundings where decisions are easy to take, the process might become cumbersome when the surroundings are less favourable and make it difficult to select good paths to reach better quality areas. This inspired us to combine two types of navigation support depending on the quality of the surroundings, one favouring flexibility (when all or most available options have good quality) and one limiting flexibility (when the environment is less favourable and the decision more complex).

2.3 Feedback Modalities

Voice-based navigation is used in most commonly available commercial systems supporting pedestrians, cyclists, runners, and in-vehicle navigation. These systems guide the user through explicit spoken commands such as “turn right” and are often accompanied by visual support on a map. Audio appears to be a suitable way to communicate navigation information as it does not require the user to carry out additional movements or shift attention too much away from the main activity itself.

Visual navigation support mainly consists of visualizing the user’s position and the path to follow on a map, usually on a smartwatch or a phone, e.g. [10]. This visualization is often provided as a fallback solution in case voice indications are missed. It also makes it possible to reassure the users at a glance that they are on the correct path. However, it requires the user to remove attention from the primary task (i.e., running or touristic walk) and explicitly look at the map.

In the context of running, haptic solutions are not commonly used to actively provide directions but rather to provide *a posteriori* warnings when the runner gets off track. Such haptic vibrations are used, for instance, by the Strava App [7] and Garmin watches [6] when the runner is further than a certain distance away from the planned path. While this kind of warning is useful, it may also easily go unnoticed or be misunderstood [11][12][13], especially if

the runner is immersed in the activity and, therefore, not attentive or receptive to the vibration signal.

From existing services and prior art, we conclude that the most suitable modality to provide directions is audio. At the same time, visual support can be used as backup and to provide decision support information to the user, and haptic feedback for alerts and to reinforce the other modalities.

3 FLEXNAV

3.1 Running Behaviours Survey

Drawing from prior art, we hypothesized that runners prefer flexible tours over fully specified ones but lack appropriate navigation support to run and explore such flexible tours in unknown places. We conducted an online survey to further understand how runners currently behave in unknown places. We used social media and snowball sampling to recruit 139 runners as respondents for this survey (69 males, 69 females, and one other, Mean age = 35.8, SD = 11.66). They were mainly based in Europe and ran on average distances of around 10 km, twice a week.

In addition to their demographic data, we asked participants about their running practices, such as the type of tours preferred (fixed/exploratory), the preparatory steps involved, their technology usage, and exploratory habits. We also asked them whether they always follow the same tours or rather tend to try out new paths. Overall, 121 out of our 139 participants stated they try to explore new paths or would like to do so (Figure 1 a). For those who already explore, we asked if and how they prepare their tours in unknown places, whether they pre-select specific areas to explore, and if so, according to which criteria. We found that 48, i.e., nearly half of the runners who already try to explore, pre-select interesting areas to explore (Figure 1 b), mainly because they are pleasant and easy to access (Figure 1 d). About two-thirds of those runners rather explore the various paths that these pre-selected areas contain instead of repeating the same loops (Figure 1 c). We also asked them whether they ever choose more than one area to explore in a single tour and found that roughly half of them do. Finally, we asked all participants if they were interested in a tool supporting the exploration of unknown places and, in that case, to indicate their email address. A total of 98 participants were interested and 36 possibly interested. Overall, the survey results thus confirmed the instinctive practice of exploring “runnable zones” among runners.

3.2 The FlexNav Concept

Our survey results show that runners pre-select and combine pleasant and easily accessible areas for exploratory running in unknown places. Aligned with these findings, we devised the FlexNav concept, which leverages this behaviour to support flexible running. FlexNav proposes running tours, linking together the best runnable zones in a neighbourhood, and provides adaptive guidance enabling the runner to follow these tours while freely exploring the runnable zones.

We define runnable zones as areas with high and consistent runnability. The runnability of an area refers to its quality in terms of the characteristics that make a place suitable for running, as also discussed in [31][32]. These include factors such as greenery,

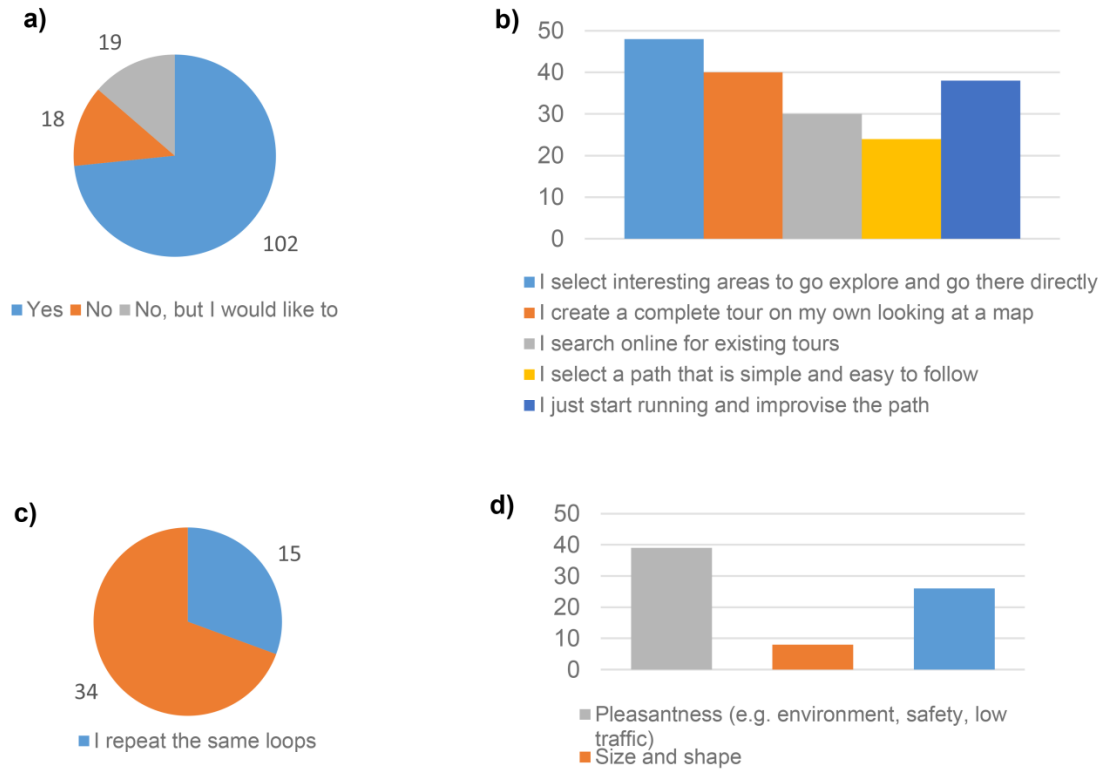


Figure 1: Running behaviour captured by our survey (figures show the number of participants who selected the corresponding options): a) Do participants run in unknown locations? b) How do participants running in unknown locations plan their tours there c) How do participants selecting areas to explore behave within these areas? (multiple choice question) d) How do they select these areas? (multiple choice question)

low-level traffic, type of road (e.g., residential street). The identification of runnable zones is based on an analysis of map data and the runnability of the paths an area contains. To determine a path's runnability, first scores for the various factors are computed and then combined as a weighted sum into the final runnability score as described in [14]. Then, based on the paths' runnability, the algorithm detailed in Annexe A.3 identifies runnable zones in the neighbourhood, i.e., areas that enable meaningful exploration that present a good overall runnability, have a significant size, and contain a significant number of alternative paths.

In some borderline cases, this algorithm will not be able to identify well delimited runnable zones in a neighbourhood. This is the case, for instance, if all the paths in a neighbourhood have roughly the same (good or bad) runnability score. In these cases, the user will be informed about the reasons why no delimited runnable zones could be identified. In case the overall runnability is good, the user can then decide to explore the whole neighbourhood freely. The runner can then still benefit from the alerts concerning the distance covered during the run and explained below: indeed, the system will notify the runner when it is time to go back to the starting point and propose the best path to get there. In case the overall runnability is bad, the system can propose to change the desired

starting point of the tour, placing it closer to a more favourable neighbourhood. Another solution discussed with the participants of our user test would be to let the user manually select and edit their own custom runnable zones to explore based on a visualisation of the runnability of the paths in the neighbourhood.

Our concept is based on the generation of running tours that connect a set of previously described runnable zones in a neighbourhood in order to approximately reach the overall target distance desired by the runner (see Figure 2). The runnable zones are selected according to their proximity, quality, and size. The connecting paths between the runnable zones are selected using the runnability characteristics explained above, while trading off (additional) distance for (better) path runnability. To estimate the overall distance of a tour including runnable zones, we define, for each runnable zone, a suggested distance to spend exploring it. To ensure that the runner's overall distance limit will not be exceeded during the run, we also define a distance limit for each zone. During the run, the system will then continuously monitor the distance already covered by the runner, notify them when reaching the suggested distance, and warn them when reaching the distance limit. Ultimately, if the runner decides to stay in the current zone beyond the distance limit, the system can dynamically modify the remainder of the tour by

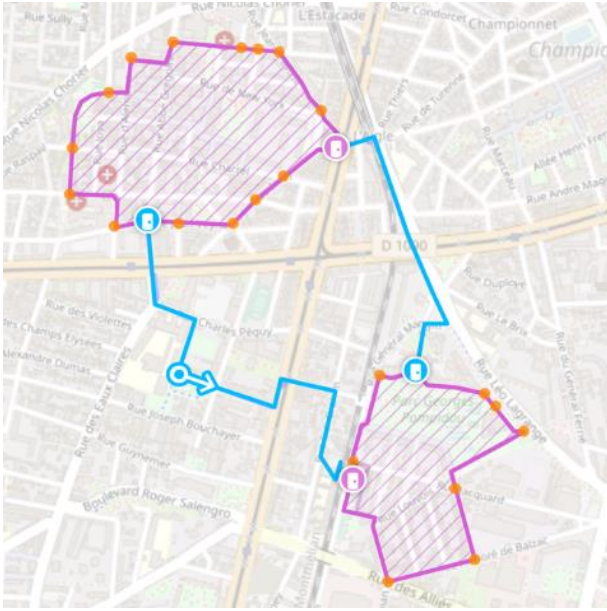


Figure 2: Illustration of our concept presenting a tour with two runnable zones (pink), their respective entries and exits (pink and blue door icons), the wrong exits (orange dots), the connecting paths (blue), and the starting point (arrow icon).

removing subsequent runnable zones to comply with the runner’s desired target distance.

During the run, we provide the runner with adaptive navigation support at various levels. The system automatically adapts this guidance level by detecting in which part of the tour the runner is currently running:

- *On the connecting paths, i.e., outside a runnable zone*, our system provides *strict* turn-by-turn navigation to efficiently and precisely guide the runner towards the next runnable zone (or the end of the tour).
- *Inside a runnable zone*, our system provides *minimal support*, thus minimizing disruption. It only provides guidance in the following situations: (1) when approaching an exit of the runnable zone, our system informs the runner whether this is the planned exit to leave the zone (and continue the tour) or a wrong exit; (2) when the runner exits the zone through a wrong exit, the runner is warned and guided back to the runnable zone. Then, when the runner leaves the runnable zone through the planned exit, the navigation support switches to the *strict mode* described above. At any time, the runner may also explicitly request navigation support to be guided to the next runnable zone.

3.3 Smartwatch Application Design

We have designed FlexNav as a smartwatch application. Our design integrates the features described above in a dedicated app combining audio, visual, and haptic feedback modalities to support navigation, building on the prior art studied in section 2.3. We use audio as the primary channel to provide the runner with all the

essential navigation support required to follow the tour confidently and without getting lost. Visual and haptic feedback modalities are used as additional channels to complement the information provided via the audio channel. Indeed, we use visual support to complement the audio support with additional information that the runner can consult on the watch screen at any time. It can be used to double-check audio instructions that might have been missed or misunderstood, or to make more informed decisions about the direction to take. Finally, we use haptic feedback, i.e., the vibrations of the watch, to notify the runner whenever indications are given through the other channels, to ensure they do not go unnoticed. For safety reasons, we limited the user input to the system to a few physical interactions with the watch’s touchscreen and physical buttons.

FlexNav thus delivers all essential navigation support via audio, and mainly via voice messages, through headphones. On the connecting path, this consists of turn-by-turn instructions at each decision point (e.g., “*Head east on Chemin des Marronniers for 200 meters*”) and alerts whenever the runner has left the correct path (“*You are off the route*”). Within the runnable zones, this consists of messages delivered when important milestones are reached on one hand, and when alerts have to be given to the runner on the other hand. The milestones are (1) entering and (2) exiting a runnable zone, as well as reaching a zone’s (3) suggested or (4) limit distance. When entering or exiting a runnable zone, the corresponding voice message also indicates respectively the suggested distance to spend in the zone or the total distance ran until that point (e.g., “*You have entered a runnable zone. The suggested distance is 1.5 km*”). The alerts given within the runnable zones consist of warnings related to wrong exits and are meant to prevent the runner from inadvertently leaving a runnable zone at such a wrong exit: a first alert, in the form of a ringing bell sound, is given to the runner when approaching a wrong exit; a second alert is given in voice format in case the runner effectively leaves the zone through a wrong exit (“*You are out of the zone*”). It is worth noting that within the runnable zones, the audio feedback is thus reduced to a minimum to limit disruption. Indeed, there is no feedback unless the runner is close to a wrong exit and gets corresponding alerts or when he or she reaches the suggested or limit distance. In contrast, the audio feedback on the connecting paths continuously provides voice messages at all decision points.

The second modality is the visual interface of the smartwatch. It consists of two screens showing the progress of the runner. The first screen shows the progress on a map, and the second one on a progress bar indicating the relative distance already covered compared to the overall tour distance. This visual interface is meant to provide supplementary and decision support information that the user is not required to consult to successfully complete the running tour.

The first, or main screen of our smartwatch application (Figure 3), displays the map of the area, with the runnable zones highlighted in pink and the paths connecting them in blue. The map also shows in grey the trace of the path already ran as well as the current position, either as a blue icon (when on the connecting paths) or as a pink icon (within the runnable zones). Turn-by-turn instructions and warning messages are displayed in addition to the corresponding audio messages on the bottom of the screen. These messages are

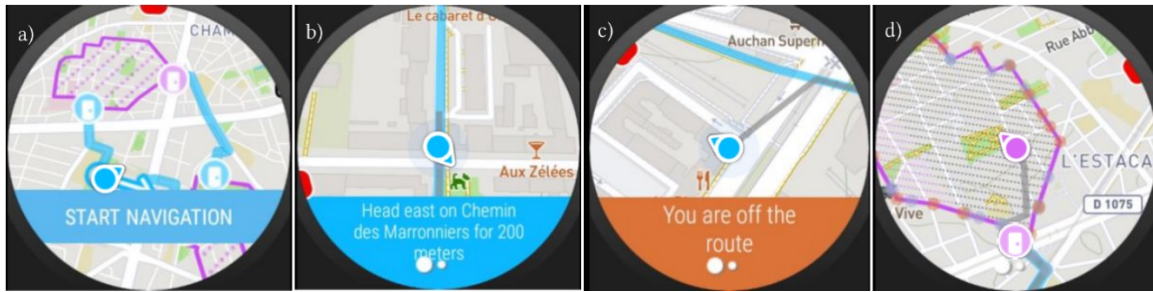


Figure 3: Main screen (map view): a) at the start, b) on a connecting path, c) when leaving the route, d) within a runnable zone

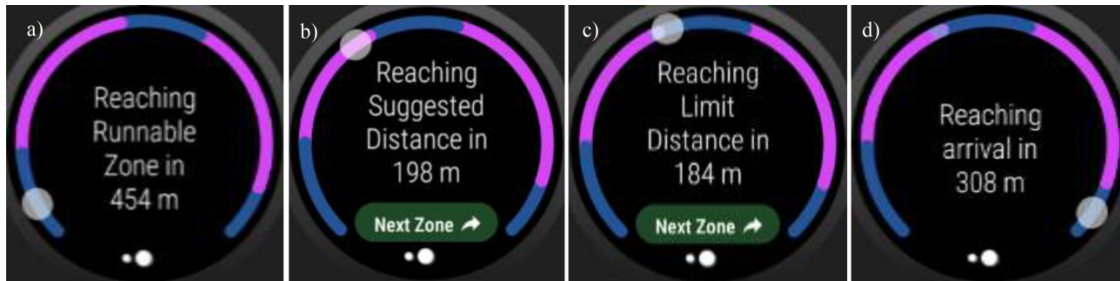


Figure 4: Second screen (progress bar): while a) on the first connecting path, b) within the first runnable zone before reaching the suggested distance, c) within the runnable zone before reaching the limit distance, d) on the last connecting path

displayed on a blue background for regular directions, and on an orange background for warnings (e.g., when leaving the right path or a runnable zone through a wrong exit). A small red oval indicates the north of the map on the periphery of the screen. We added this north indicator to help the runner interpret navigation instructions that refer to compass directions such as “head north-east”. Finally, the map also highlights the starting point and direction of the tour, the runnable zone entries (as pink gates), correct exits (as blue gates) and all the wrong exits of the zones (as orange dots).

The zoom level of the map is automatically adjusted to the current running context. When following a connecting path or running at the border of a runnable zone (close to an exit), the zoom level is increased to enable the runner to better appreciate the immediate environment and follow the instructions given by the system. Within the zones, the zoom level is decreased to help the runner grasp the extension of the whole runnable zone and decide which parts to explore. At any time, the runner can change this zoom level by double-tapping the touchscreen with one or two fingers to zoom in or out, respectively, or explore the map by panning.

The second screen (Figure 4) displays information about the progress made by the runner. The runner can access this screen by pressing a (physical) button on the watch. This screen contains an arc-shaped progress bar and a white dot representing the runner’s current position and progress with respect to the overall distance to be covered. The progress bar shows connecting paths in blue and runnable zones in pink, corresponding to the colours used in the first screen. The length of the different segments corresponds to the relative distance they cover. At a glance, the runner can thus understand in which segment they are currently running and how much distance they have already covered. Furthermore, the

centre of this screen displays the remaining distance to reach the next milestone of the tour. Within a runnable zone, this screen also contains a button enabling the runner to ask for turn-by-turn guidance to the next runnable zone (or the end of the tour), which the runner can use at any time.

Finally, when the runner reaches a major milestone of the tour, e.g., entering or exiting a runnable zone or finishing the tour, a visual full-screen notification message (Figure 5) is overlaid on top of the other screens.

3.4 Implementation

We have implemented a prototype of FlexNav on a Suunto 7 sport smartwatch [15], which runs the Wear OS operating system. Our choice was guided by the fact that (1) this is the kind of sport watch runners already use [22], (2) its large (50 mm- 454x454 pixels) screen makes it possible to display a map and directions in a readable way, and (3) it enabled us to program and install a dedicated application implementing our concept (using standard Java programming).

We developed our prototype as a Wear OS app on Android Studio [16]. We integrated it with Mapbox SDK [17] to support the display of maps and the retrieval of turn-by-turn navigation instructions. The watch was paired with headphones, using Bluetooth, to play audio instructions to the runner. However, one drawback of this smartwatch was that it didn’t come with a built-in cellular network connection. Therefore, we had to equip our participants with smartphones paired with the watch in order to enable the communication with the various required online APIs that enable (1) to fetch all the available tour itineraries, (2) to retrieve the corresponding textual navigation instructions (Mapbox API [17]), (3) to convert these text

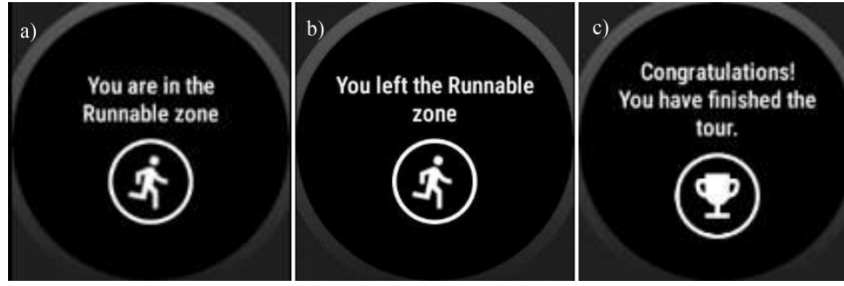


Figure 5: Full-screen messages: a) when entering a runnable zone, b) when exiting a runnable zone, c) when finishing the tour

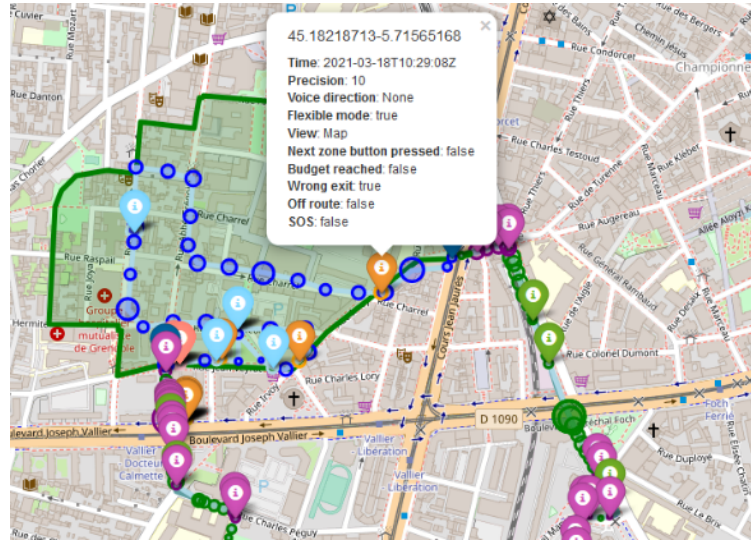


Figure 6: View of the monitoring interface showing a participant's progress through timestamped events at corresponding locations.

instructions to speech (Text-to-Speech API [18]), and (4) to update our backend about the runner's progress.

The updates sent to the backend consisted of timestamped information (Figure 6), including the GPS location and precision, the screen currently displayed on the watch (map or progress bar), the navigation instructions given, the warnings issued (i.e., about the suggested distance reached, being off route, or approaching a wrong exit) and whether the runner had requested to be guided to the next runnable zone (or to the end of the tour). This information was then stored in our database and made accessible through a map-based interface for the research team to monitor the participant's progress in real-time (see Appendix 1 and 2 for more details).

4 EVALUATION

To evaluate our system, we conducted an experiment in a medium-sized French city. We manually crafted a tour with two runnable zones so that the tour corresponded exactly to our needs (see below). Participants had to complete the tour using our FlexNav smartwatch application prototype. Our experiment was designed to investigate

if our concept satisfies the needs of runners to run in unknown locations. More specifically, we want to answer the following research questions:

- How can runners' needs and preferences for running tours in unknown locations be met? Do runnable zones linked by connecting paths together with adaptive navigation support better satisfy these needs and preferences than fully specified tours together with strict turn-by-turn guidance?
- What are runners preferences regarding having one or several runnable zones with the same or different characteristics in a single tour?
- What level of guidance provides better support for exploration and spatial learning to runners, runnable zones with minimal guidance or a fully specified tour with turn by turn guidance?

Initially, we had planned to have the participants run a second fully specified test tour with strict turn-by-turn guidance for comparison. However, we refrained from doing so to limit the participants' effort and fatigue and the duration of the experiment.

Moreover, we considered that through the connecting path segments of our test tour, our participants already experienced a fixed path with turn-by-turn guidance over a sufficient distance.

Thus, we crafted one test tour intending to obtain a total distance of roughly 5 km, including two runnable zones of roughly the same size in different neighbourhoods (Figure 7). We included two zones in order to understand how runners react to multiple runnable zones with different characteristics. Both zones were a mix of green and low-traffic areas. The first zone contained a large park and adjacent residential living streets, and the second zone had a more urban setting, including a few small green spaces and streets with shops and restaurants. Both zones were relatively flat to exclude elevation difficulty. They also contained a significant set of alternative paths to enable meaningful exploration. As our focus was on evaluating interest in the runnable zones, we defined large enough zones, limiting at the same time the distance covered by the connecting paths. Overall, our tour contained three connecting paths covering a bit more than 2 km in total, and two runnable zones with a suggested distance of 1.5 km each. During the experiment, we collected feedback about our participants' experiences.

4.1 Participants

To recruit participants for this experiment, we sent out a new survey to the participants of our first survey who resided in our city and had stated they were interested in a tool supporting exploratory running. We asked them to indicate how well they knew the different sectors of that city. This enabled us to prepare the test tour in a less known part of the city and pre-select participants unfamiliar with that test location for our user experiment. We finally recruited 26 participants (14 male and 12 female), aged from 18 to 64 years (Mean age = 38.5, SD = 12.2). A total of 24 participants had stated in the first survey that they wanted to be able to explore unknown places, but only 22 had already engaged in exploratory running. Among those 22, to prepare exploratory running, 6 usually searched for available traces from other runners and followed them, 14 usually prepared their exploratory runs by selecting an interesting area on the map, such as a park or a similar green area, as well as a path to reach that area. Once in that area, they stated that they explored it freely. Most of our participants ran regularly, once or twice a week, while 7 participants were occasional runners running less than once a week. Concerning the distance, 13 participants usually run between 5-10 km, 9 usually run more than 10 km, and 4 less than 5 km. Most, except 4, were usually running in urban environments. To compensate them for their efforts in participating in our experiment, each participant received a voucher equivalent to 25 USD.

4.2 Measures

To answer our research questions, during our experiment we collected measures about (1) the spatial learning of the environment, (2) the appreciation of our FlexNav concept, and (3) a subjective comparison of FlexNav with a fully Turn-by-Turn (TbT) based system.

To measure the spatial learning of the participants in runnable zones compared to connecting paths, we asked our participants to place on a map 12 intersections they had crossed when running our

tour. Among these 12 intersections, 6 were from connecting paths and 6 from runnable zones. We pre-selected 2 intersections from each of the 3 connecting paths (as all the participants had to cross them during their run), and we randomly selected 3 intersections from each of the 2 runnable zones during the experiment, considering only those that the participant had crossed during the run. We showed the participant a 360-degree Google Street view of each of these intersections on a laptop, asking them to indicate its location on a printed map. We acknowledge that the varying characteristics of the intersections, e.g., the presence of landmarks, can play a role in its identification. However, none of our intersections had outstanding landmarks. They were mostly street corners with houses, sometimes commercial spots, parks, or big boulevards.

We collected the participants' opinions about our concept and their overall experience through a semi-structured interview. This interview contained a mix of five-item Likert-scale questions, followed by their verbal explanation for the given ratings, and open-ended questions. Finally, similarly to the evaluation method used by Komminos et al. [9], we asked the participants to subjectively compare our FlexNav system with a fully TbT system (see post-test questionnaire in Supplementary Material). Nearly none of our participants had already experienced any of the commercial systems available for running ([5][6][7] or similar) or used TbT guidance during running. However, they all had experienced TbT guidance in our user test when following the connecting paths with our system. We clarified with each participant that they should compare their overall experience on our test tour (combining connecting paths and zones and different modes of navigation support) against a fully specified tour with TbT navigation (as provided on the connecting path segments).

4.3 Procedure

Before the experiment, we sent the participants documents providing a general overview of the experiment, the data to be collected, its storage and processing, as well as the consent form, which they signed and sent back to us. For the experiment, we welcomed each participant at the starting point of the tour. We sat on a park bench to describe the experiment and we equipped the participants with the prototype, i.e., the smartwatch, the headphones, and the smartphone, which they carried in a pocket or on an armband during the run. We then familiarized the participant with our system on a small training tour crafted with a small park as the runnable zone. In the training tour, the participant experienced all the functionalities of the system.

We then instructed the participant to run the test tour with our system and stressed that completion time was not an evaluation metric. We asked them to explore the runnable zones as desired, if possible, reaching the suggested distance. We pointed out that the GPS data were not 100% accurate, sometimes resulting in erroneous navigation instructions that they were supposed to handle independently. We provided a contact to call in case of emergency, e.g., getting lost or having unexpected errors or accidents. Finally, we directed the participant to the start of the tour. We then recorded the confounding variables such as the weather, day, and time. After the run, we asked the participant to complete the questionnaire

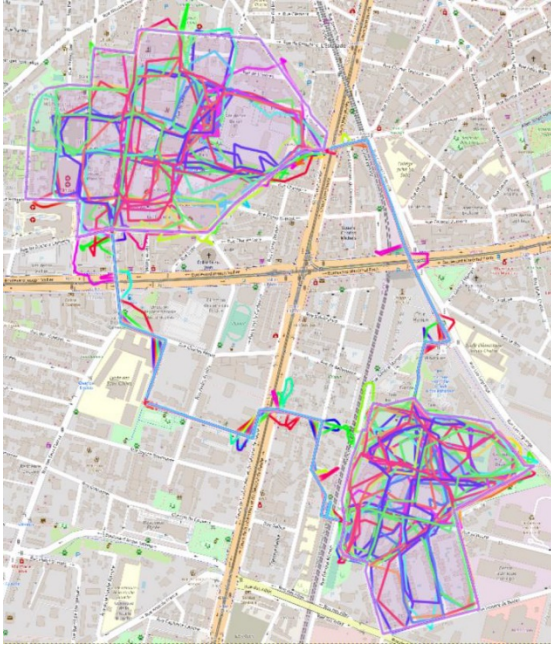


Figure 7: Visualisation of the running traces of our 26 participants.

described in 5.2 and debriefed through a semi-structured interview focusing on the overall experience.

During this experiment, we took all necessary COVID precautions. The experimenters and participants wore masks pre- and post-run, i.e., while giving explanations or collecting post-experiment data. All components of the prototype were sanitized before and after each run.

5 RESULTS

All participants successfully completed the experiment (Figure 7) without any significant system (or user) failure, and no participant contacted the experimenters for assistance. A total of 17 experiments were conducted in French, 8 in English, and 1 in Spanish. The experiments took place in variable weather conditions (2 to 19 degrees Celsius, rainy to sunny weather) and at different times of the day. However, we observed no effect of these confounding variables on the results. In the following, we will first summarize the quantitative results before discussing our qualitative findings in more detail.

5.1 Quantitative Data

5.1.1 Spatial Learning evaluation. We tested the participants' recollection of 12 intersections picked from all the intersections they crossed during their itinerary. We asked them to place each of these intersections on a map, either approximately, i.e., selecting only the area where they thought the intersection was located (i.e., the corresponding connecting path or runnable zone), or, if possible, precisely, i.e., selecting one specific intersection. It is worth recalling that the neighbourhood within each of the two runnable zones was relatively homogeneous and different from the other zone and

the connecting paths. With the approximate mapping, the average correct rate reached 62%, with 58% for intersections in connecting paths and 65% for intersections in runnable zones (shown in Figure 8). With the precise mapping, the performance of the participants varied dramatically, ranging from 0 to 12 correctly placed intersections, with an average correct rate of 36% per participant (49% in connecting path and 23% in runnable zones). Considering that the runnable zones contained many more intersections than the connecting paths (making the correct placement more difficult), we also computed a weighted precise mapping rate, weighting the number of correctly placed intersections with the proportion of intersections located in each zone or connecting path. This resulted in a correct rate of 19% for the intersections located on the connecting paths and 45% for the intersections within the runnable zones (Figure 8). These results suggest a higher recognition rate in the zones, answering our third research question: less navigation support encourages more spatial learning. However, there are certain considerations to be made, as discussed in the qualitative section.

5.1.2 Overall appreciation of the concept. During the debrief interview, we asked the participants questions on a five-item Likert-scale to obtain quantitative feedback about their overall appreciation of the FlexNav system (Figure 9). Over 90% of the participants liked the concept and the amount of navigation support it provides in both modes (minimal in zones and strict TbT guidance on connecting paths). Over 80% stated that the system gave no unnecessary instructions. Over 80% also liked the idea of having more than one runnable zone in a tour and felt that they discovered more about their surroundings in the runnable zones than in the connecting paths. All of them stated that they would like to use the system in the future to explore unknown places. Participants also responded to a number of open-ended questions that we share and discuss in the Qualitative results section.

5.1.3 Comparing FlexNav with full Turn-by-Turn guidance. The final section of our semi-structured interview was designed to compare our system with a system providing fully specified tours with TbT navigation. Most participants had never used such TbT guidance for running before, but they could easily extrapolate from their experience with our system on the connecting paths. For each question, the participants selected an answer from three options: *FlexNav*, *TbT*, and *No difference*.

Over 80% of the participants found our FlexNav system more enjoyable in terms of navigation support. They stated that it gave them the ability to focus more on the environment and that it was better in supporting exploration (Figure 10). 50% of the participants found TbT guidance more disruptive, against less than 20% for FlexNav. Finally, the participants rated both systems roughly equal on ease of understanding, stressfulness, and safety.

5.2 Qualitative Data

We analysed the qualitative data we collected using thematic analysis [19]. First, all audio recordings were transcribed and then deductively axially coded [20] based on the interview questions. This was done individually by all the authors and then discussed and agreed upon. Next, the data was inductively coded for repeated themes, e.g., zone characteristics, participant motivations, sense of

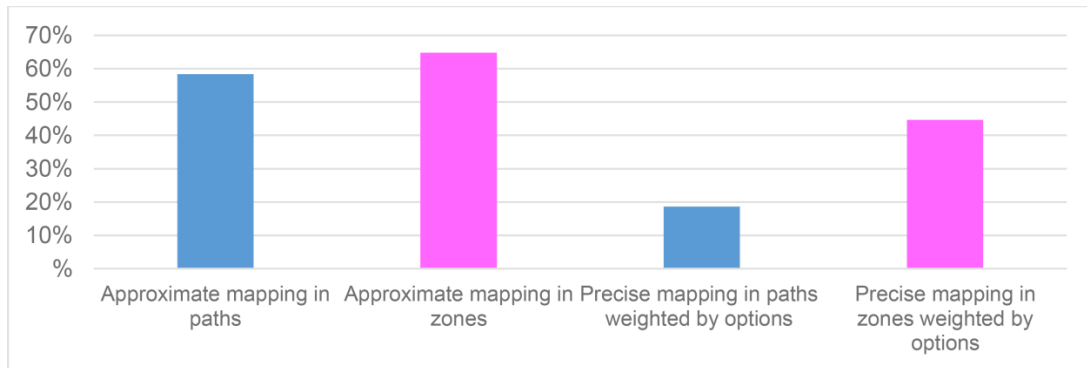


Figure 8: Proportion of correctly mapped intersections (in paths and zones): approximate mapping and precise mapping weighted by the total number of intersections located in the connecting paths or runnable zones.

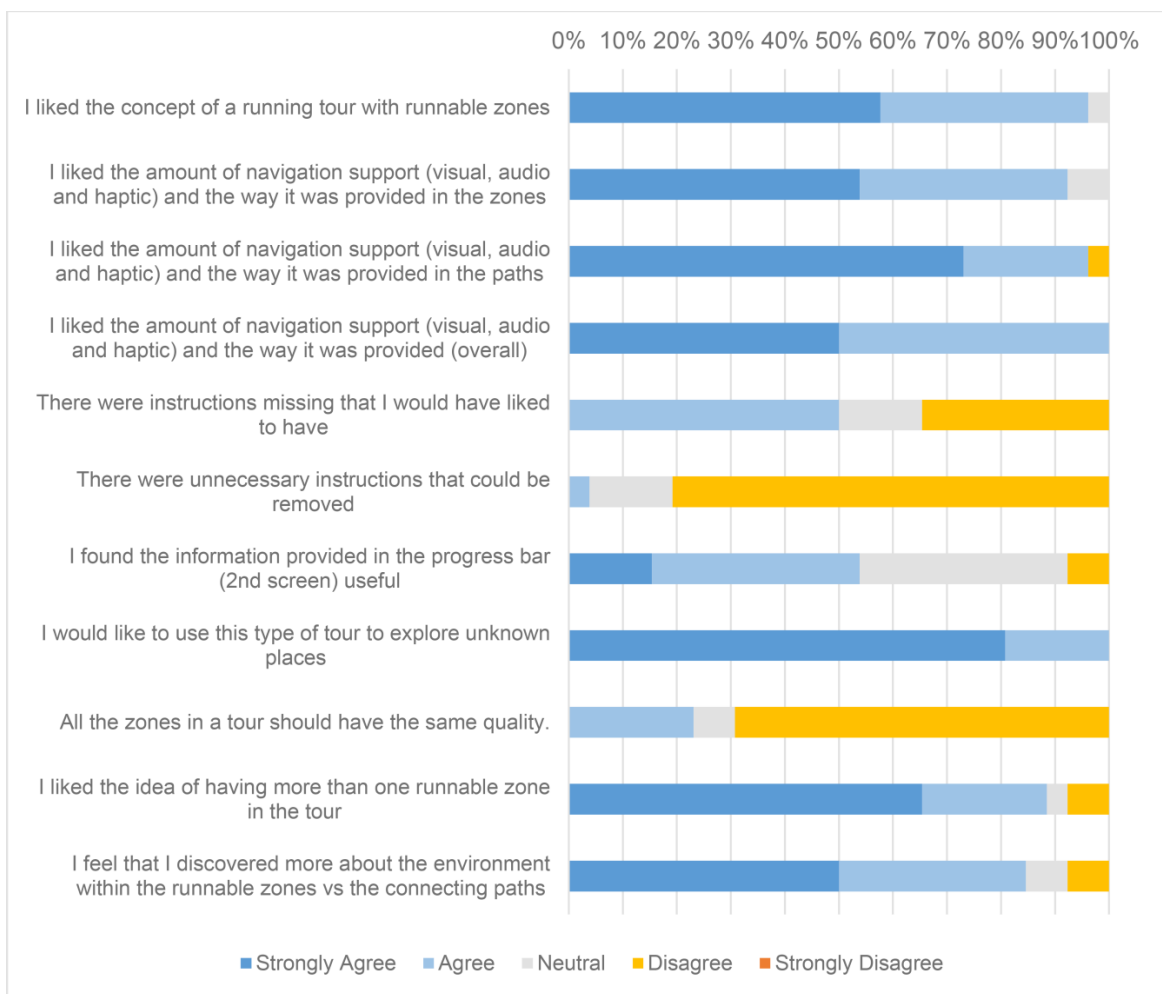


Figure 9: Overall appreciation of the FlexNav concept. (Note that there were two negative questions, namely “There were unnecessary instructions that could have been removed” and “All the zones in a tour should have the same quality”).

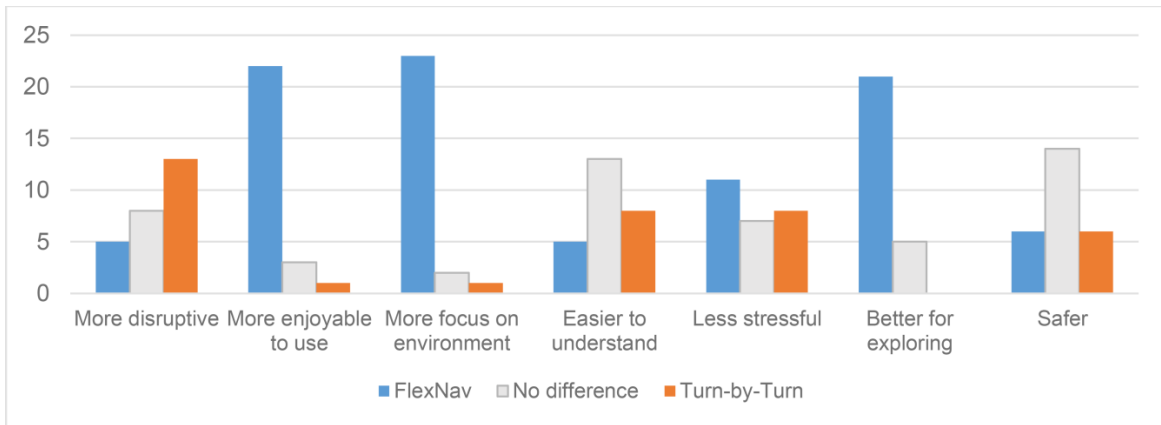


Figure 10: Comparison of FlexNav versus Turn-by-Turn guidance showing the number of participants who chose each option

freedom, context of use, area discovery, in a collaborative process. Finally, we selected the frequent instances mentioned by the participants as the main categories and sub-categories as this indicated a significant dimension of common ground.

Overall, all participants liked our concept and found the system valuable for exploration. They mentioned that they would like to use it to discover unknown locations, both on vacation as well as in their home city. They said it would enable them “to go out of [their] comfort zone or not [follow] the same path every time” (P4). They found that our system would help them save the time required for preparing the runs while providing a feeling of security.

5.2.1 Appreciation of runnable zones. When defining the runnable zones in our tour, we considered greenery and low-level traffic as desirable core characteristics, in line with existing algorithms discussed in the Related Work section. The runnable zones contained quiet residential neighbourhoods and parks, whereas the connecting paths also comprised wide avenues or streets with higher traffic levels. The participants’ expectations from a zone matched these characteristics. They frequently mentioned “parks, few people, little traffic” (P23) as key elements that clearly differentiated the runnable zones and made them more enjoyable. They considered the zones as “more nature-like” (P8) and as a safe and relaxing space for running without disturbing crowds or noises:

“First, it is safe to run there [in the runnable zones]. Also, there are not too many people. Like there are no restaurants, shops, or things that make the zone to be crowded. And it is greener, and there is not much noise, so we could relax.” (P4)

Figure 7 shows that participants really explored the interior of the zones and especially the parks they contained. They avoided running on the zone borders and explored less the parts that didn’t have green spaces or interesting architecture (e.g., the streets in the south of the first runnable zone with a more industrial look and less nature). This behaviour validated the appropriateness of zone characteristics in practice. A few participants pointed out the difficulty of reaching the suggested distance within a zone, finding that “(to) make the distance you have to loop around a bit within the zone, which can be a bit boring” (P3). This may be related to the limited size of the zones we crafted.

5.2.2 Adaptive navigation and guidance modalities. Participants clearly liked the adaptation of the navigation support to the different parts of the tour, i.e., from strict TbT support on the connecting paths to minimal support in the runnable zones. Due to the nature of the connecting paths, which usually had higher traffic levels and were more crowded, participants found the stricter navigation support more appropriate and paid attention to the instructions. In the zones, in contrast, the participants felt more relaxed, relied less on the navigation support, and “looked less at the screen” (P17). They liked the absence of regular instructions:

“No, there wasn’t too much information at all. Even when I got to the first zone, I thought to myself, it’s been a long time since I heard anything. That was good.” (P10)

At the same time, some participants had doubts about the usefulness of the wrong exit alerts given at the border of the runnable zones. They did not like receiving these regular warnings when running on the border of the zones and coming close to wrong exits. They even mentioned a conscious effort to avoid running across wrong exits to avoid these alerts, which increased their mental load. They stated they “would have liked fewer alerts, e.g., just when I really leave the area” (P5). In consequence, as we can also see in Figure 7, participants usually rather explored the paths within the zones and kept away from the borders.

The multi-modal interface of our system suited the participants’ needs. They found that the “three information modalities were interesting” (P17), “very easy to use” (P8), and gave the right amount of information. However, some participants also stated that they were not using our system and its modalities optimally because they were using it for the first time, but would be able to use it more efficiently the next time:

“For the first time ... I was a little too much on the watch, especially at the beginning, trying to see where I could go without leaving the zone. If I reuse it in the future, I could look at the environment more.” (P2).

Indeed, when analysing their ratings comparing our system (i.e., a new and more elaborate concept with different levels of navigation support) to simple turn-by-turn navigation (which most of them had already used for other activities), one has to consider that this was their first experience with our system.

5.2.3 Freedom vs. Responsibility. Participants liked the shift in power, from being passively guided on the connecting paths to being responsible for looking and selecting the paths to explore within the runnable zones. By exploring the zones freely, they indulged in the pleasure of serendipitous discovery without the pressure of pre-planning in advance. The participants had the confidence to disconnect and felt encouraged to discover more of the environment when running in the zones:

"When you run in a free zone, you forget a little about the constraints of not knowing the place. It gives a certain confidence that is pleasant. You know that as long as you don't go out of this zone, you are calm, you can disconnect." (P22)

We can thus answer the first part of our third research question, as participants considered that our system provides easier and better exploration support than full TbT navigation support. Furthermore, participants even perceived reaching a runnable zone as an additional motivation. Indeed, each runnable zone represented a goal and an opportunity for new discoveries. Hence, viewing it as an incentive, they also liked having more than one runnable zone in the tour:

"It was good to change zones. It sets goals, I liked it. I said to myself, I have finished one zone and we will go to another one with even more discoveries. That's what I liked." (P13)

While FlexNav gave the runners freedom to explore, it was also considered as mentally more demanding than a fully specified tour with turn-by-turn guidance. Indeed, some participants explicitly attributed this mental load to the constant decision-making required to choose the path to follow within the zones. Others mentioned that the absence of guidance, i.e., the system *"doesn't speak at all in the flexible zones"* (P14), generated additional stress compared to TbT navigation support:

"FlexNav is more disturbing in the sense that it requires more thought, attention, concentration and more focus on the course (one has to take decisions!) whereas with TbT there are no questions; one is constantly guided." (P6)

5.2.4 Spatial Learning. The impact of navigation support on learning about the environment has been previously investigated in the literature for pedestrians [21][36], but not yet for runners. While most participants stated that, during the run, they were not really focused on learning and remembering the environment, in general they all agreed that they looked more at the environment within the zones when running freely. This may be expected, as, at each intersection, they had to select the path to follow. Accordingly, as discussed above, we observed slightly better learning within the zones as opposed to the connecting paths in participants' spatial learning measures (Figure 8):

"With FlexNav, one is more motivated because it is you who makes the choice, so you look at the environment more to decide where it looks nicer and where you want to explore ... On the connecting paths where I was guided, so I did not look so much at the buildings/surroundings." (P9)

However, the learning behaviour differed from person to person. For example, some participants remembered particular features in the environment, such as *"a railway bridge [they] had never seen before"* (P3). Others rather explicitly looked at street signs or remembered the street names mentioned in the TbT speech

directions. Still, others tended to better remember more complex intersections or intersections where they faced specific navigation issues:

"This one [intersection], I remember: when following the [voice] indications, I hesitated with another intersection which was just before ... a yes ... it's here [on the map]" (P25)

Overall, there were several person-specific factors explaining why participants remembered one intersection rather than another, which made the analysis of the quantitative results more difficult. This variability in learning behaviour prevents a strong answer to our third research question, namely, whether runners learn more about the environment when getting minimal guidance (such as in the zones) rather than when getting turn-by-turn guidance (such as on the connecting paths).

6 DISCUSSION

Below we discuss our findings with a focus on (1) the runnable zones and their characteristics and (2) the feedback modalities and the level of support provided by our system.

6.1 The subjective nature of the Runnable Zones

We confirmed that the characteristics we considered important for designating a place as suitable for running were indeed also relevant for the participants. Interestingly, most participants did not find it necessary that all runnable zones within a tour should have the same characteristics and runnability. They rather stated that having runnable zones with different characteristics could make the overall tour even more enjoyable: *"if it consists only of green spaces, it could also get a little bit boring"* (P26). Beyond the notion of just being "runnable", the participants suggested that the system could also help runners *"discover characteristic places like interesting buildings in a city"* (P21) through these zones. At the same time, they acknowledged and accepted that the quality of the runnable zones will necessarily vary and depends on what is available in the area where they want to run.

We heard varying opinions regarding the optimal size of the zones. While some participants preferred to have bigger zones to avoid *"going around in circles"* (P17), others agreed on rather having a bigger number of smaller zones as a way to keep the overall tour interesting. Most participants found that the number of zones we selected was appropriate for the length of the tour they experienced. They believed that this number should be adapted depending on the target distance. The participants' interest in having multiple zones with different characteristics within a tour provides answers to our second research question.

The concept of runnable zones is designed to give runners more control and agency over where they want to run, letting them choose the concrete paths they explore. Allowing runners to further customize the definition of the zones would be an effective way to cater to the different interests mentioned by participants with respect to the relevant characteristics of a zone. Indeed, participants mentioned that the zones could be personalized based on their individual preferences, chosen from a zone library, or even edited by the runner, moving the *"points on the border of the zone"* (P25) to make them expandable to better suit the desired running distance.

They also imagined creating their own tour and “*defining their own zones*” from scratch “*and upload it to the app*” (P6) to benefit from the adaptive navigation while running their tour. These suggestions require runners to play an active role in crafting and planning tours. The trade-off of time invested by the runners in manual crafting and personalization of tours versus automatic generation should be carefully considered. Overall, the homogeneity of an area and the creation of zones based on one or a set of characteristics can inform the design of future tools that support exploration in other domains, such as tourism.

6.2 The right level of support

Running is an intensive physical activity that requires runners to be focused on their environment while receiving and following navigation instructions. To overcome this challenge, we designed our system to provide several complementary feedback modes. Most participants liked this and found it very effective. For instance, P9 pointed out that even though she forgot her glasses and could not read the messages or see the map in detail, she was nevertheless able to use the systems all the same, showing that “*one does not need to look at much detail to be able to use this system and follow the tour*” (P9). However, a few participants found that “*the three (feedback modes) at the same time might be too much*” (P7) and suggested that users should be able to choose the feedback mode(s) they prefer. This suggestion could indeed be easily addressed and implemented.

While we have minimized the instructions within the runnable zones, some participants preferred to have even fewer instructions. They complained about “*too many vibrating alerts*” (P15) and asked for fewer alerts on the border on the zone, to be notified only “*when (they) really leave the zone and not when approaching the border*” (P5). A few mentioned that the system should bring them directly to the centre of the zone instead of its entry to avoid being close to the borders and thus receiving the corresponding warnings. Others would have liked to get suggestions on how to reach the most interesting areas within the zone itself. While we see this as an interesting opportunity, our aim is rather to reduce disruptions and give runners the possibility to explore and discover the areas more freely. Every additional instruction proposed by the runners would need careful integration into our system to avoid compromising this core value. One possibility would be to enable users to actively request such information.

The results discussed above show that people prefer our concept of running tours with adaptive navigation support. This favourably answers our first research question. Runners find turn-by-turn navigation more disruptive than the minimal guidance provided in the runnable zones. Indeed, considering this, most found that FlexNav is better suited for exploration and allows them to focus more on the environment. However, the subtle balance between guidance and exploration should be carefully maintained in the design of systems like ours. Allowing for a personalisation of the desired amount and type of navigation support can be beneficial in future systems.

7 LIMITATIONS

Sensor inaccuracies. Our system relies on sensors that may be inaccurate due to environmental conditions, such as the buildings in the

area or the weather. GPS inaccuracies caused delayed instructions and false warnings during our tests. However, they did not prevent participants from following the instructions and completing the running tour independently. Besides the GPS, the compass feature on the UI (based on the smartwatch sensor) was also often slow and inaccurate. While only a few participants mentioned having tried this feature, they reported having problems appreciating its usefulness.

System errors. Our system relied on internet connection to fetch the audio instructions for the tour. For two of the 26 participants, some audio instructions were not given due to what we believe were connection problems during a part of the tour. In those cases, the participants coped with the problem and relied on the visual instructions and the haptic feedback to successfully follow the tour and continue the experiment. Again, this illustrates the robustness of multi-modal feedback.

Personalization. As previously discussed, many aspects of our systems could be personalized to match the specific interests of each user. This applies to runnable zone selection as well as feedback modes. We had considered this aspect in the initial design of the concept, where we discussed the possibility of letting people customise certain features. However, it was not possible for us to implement such customisation in the prototype we used in our experiment.

8 CONCLUSION

In this paper, we present FlexNav, a system that generates exploratory running tours combining runnable zones with connecting paths and provides adaptive navigation support to follow these tours. It uses turn-by-turn guidance to guide the runner efficiently from one running zone to the next and lets the runner explore these zones freely. Our system conforms to the natural exploratory behaviour of runners, as identified in our initial survey results. We described the design of FlexNav and the implementation of a prototype. Our system was tested and evaluated by 26 runners who considered it more appropriate for exploring unknown places than fully specified running tours with turn-by-turn guidance. The participants liked the idea of runnable zones, agreed with our core characteristics to define them, and appreciated the multi-modal and adaptive navigation support FlexNav provides. The findings furthermore highlight the subjective nature of the runnable zones and the desire for personalizing them as well as the levels and modalities of the navigation support provided. Our research adds knowledge on flexible pedestrian exploration as well as corresponding novel adaptive navigation systems.

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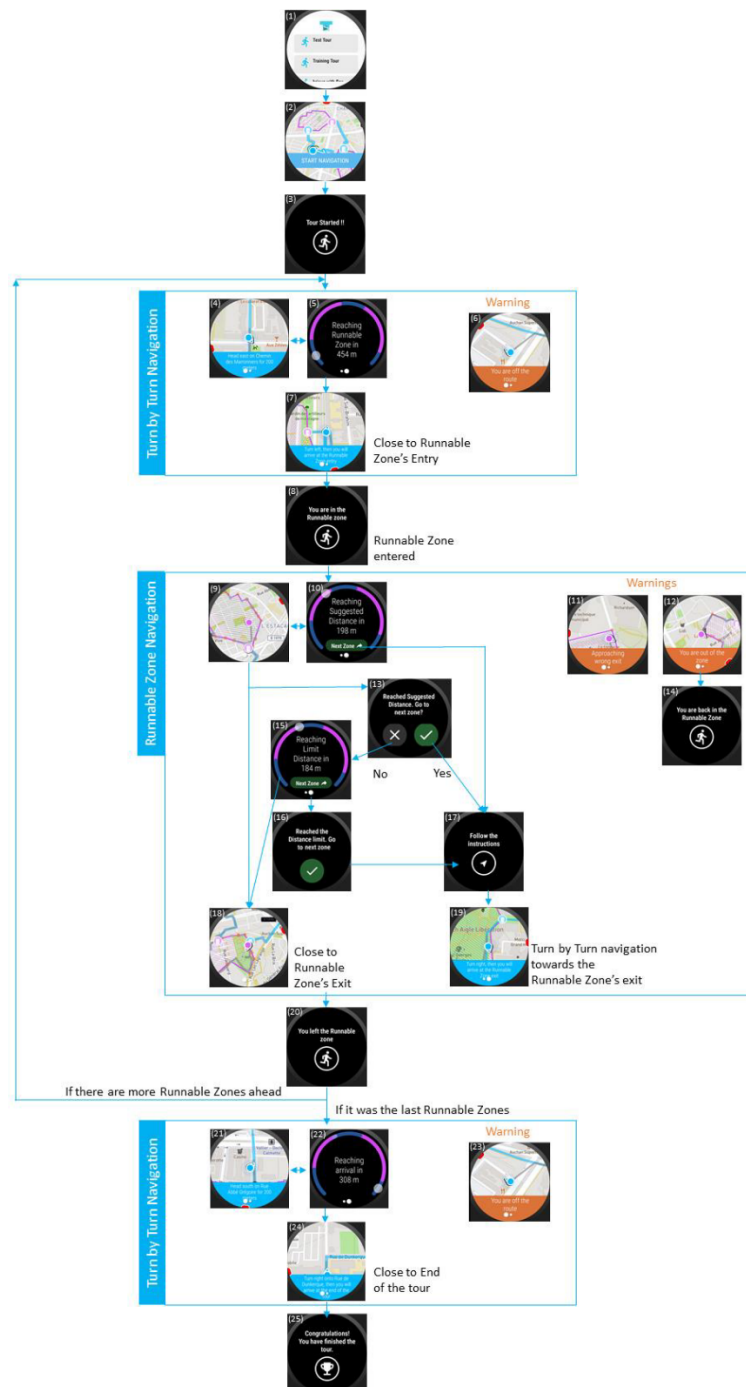
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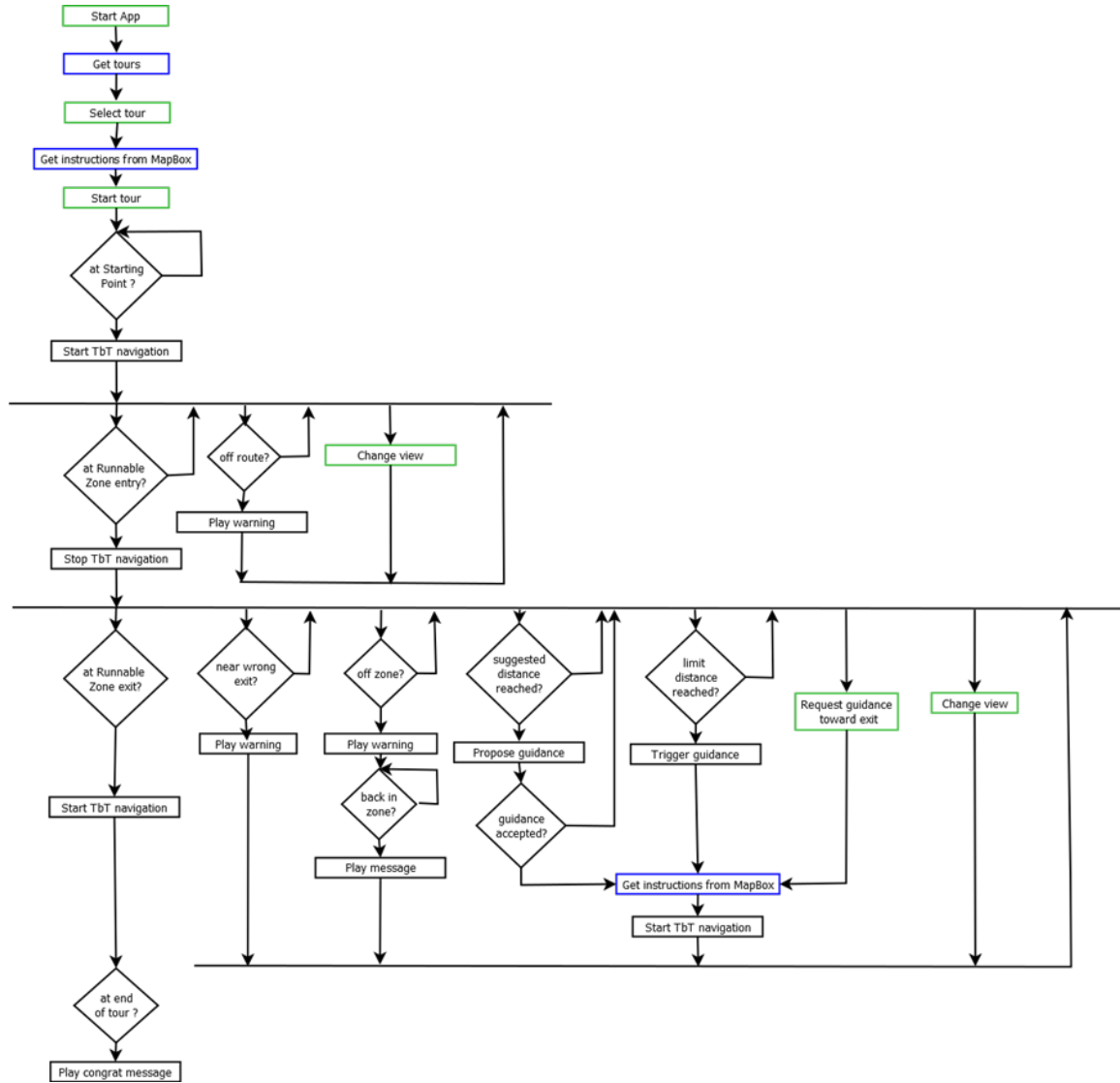
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A APPENDICES

A UI FLOWCHART



B FUNCTIONAL FLOWCHART



C THE ALGORITHM USED TO IDENTIFY RUNNABLE ZONES

Here, we provide a high-level description of the algorithm allowing to identify runnable zones from a graph representing a geographic neighbourhood. The graph used by the algorithm is initially constructed from available map data, e.g., OpenStreetMap. The graph edges represent the paths that exist in the geographic neighbourhood, and the nodes represent the intersections between these paths. Each edge of the graph gets assigned a runnability score corresponding to its (average) quality for running. This score is computed considering the characteristics of the path represented by the edge (pedestrian path or living street, for instance) and the quality of its environment (its closeness to green spaces on one hand and highways on the other hand, for instance) as described in [14]. Furthermore, the graph contains information about its geometry, such as the localization of the paths and intersections and the

angles between the paths. As explained below, in a first step, the algorithm then uses thresholding and pre-processing to identify candidate runnable zones from this graph. In a second step, it then rejects insignificant, inhomogeneous, and in-sufficiently flexible candidate runnable zones to finally consider only the best runnable zones for inclusion as runnable zones into running tours. In a first step, the algorithm identifies candidate runnable zones as follows:

- 1) Thresholding is used to distinguish three types of edges in the graph
 - a) *desirable edges*, i.e., edges that are well suited for running and that have a score above a desired threshold runnability value S_1 ;
 - b) *unacceptable edges*, i.e., edges that are not well suited for running and that have a score below an unacceptable threshold runnability value S_2 , with $S_2 < S_1$;

- c) *acceptable edges*, i.e., edges that have a score between S_1 and S_2 .
- 2) The algorithm then blends out all acceptable and unacceptable edges to identify all subgraphs exclusively constituted by connected desirable edges. These resulting subgraphs are only connected one to another by edges that have a lower than desirable runnability score.
- 3) For each subgraph, the algorithm then identifies its largest inner tour, i.e., the sequence of desirable edges in this subgraph that maximizes the surface enclosed in the corresponding geographic area. The geographic area delimited by this boundary constitutes then a candidate runnable zone, and all the edges this area contains constitute the possible paths belonging to this candidate runnable zone. N.B. This area may still contain also edges that have a lower than desirable runnability score
- 4) For each subgraph, the algorithm then identifies its largest inner tour, i.e., the sequence of desirable edges in this subgraph that maximizes the surface enclosed in the corresponding geographic area. The geographic area delimited by this boundary constitutes then a candidate runnable zone, and all the edges this area contains constitute the possible paths belonging to this candidate runnable zone. N.B. This area may still contain also edges that have a lower than desirable runnability score

In a second step the algorithm then rejects insignificant, insufficiently runnable, and insufficiently flexible candidate runnable zones as follows:

- 1) It rejects insignificant candidate runnable zones that are not worthwhile exploring because they do not cover a significant cumulated distance or geographic area. This sub-step applies threshold values for the minimal required cumulated distance and the minimal required geographic surface for a runnable zone.
- 2) It rejects in-sufficiently runnable candidate runnable zones, i.e., zones with an average runnability score of all the edges it contains that is below a minimum required runnability score.
- 3) It rejects insufficiently flexible candidate runnable zones, i.e., zones that do not enable a significant variety of alternative paths. This sub-step uses a threshold value for the minimal number of alternatives required for a runnable zone.

The threshold runnability values used in the first step can be adjusted to increase/decrease the number of candidate runnable zones to consider in the second step.

The threshold values used in the second step can also be adjusted to limit the number of runnable zones to consider for inclusion into the proposed running tours, e.g., according to the characteristics of all the candidate runnable zones found in the first step, for instance, by using the median values of these candidate zones.